

# Antreas Antoniou

Machine Learning Research Scientist & Engineer

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I design and train learning systems, diagnose failures and optimise them for deployment. I also build the cloud infrastructure, evaluation frameworks and software needed to take them from research to working systems.

## Technical Skills

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**Model development:** pretraining, LoRA/QLoRA, full fine-tuning, distillation, meta-learning, multimodal world models, embeddings, persistent memory and retrieval.

**Evaluation & delivery:** experiment design, benchmarking, training diagnosis, synthetic-data pipelines and on-device optimisation; Swift, CoreML.

**Training & infrastructure:** Python, PyTorch, TensorFlow, FSDP, Accelerate, W&B, Hugging Face; GCP/GKE, Docker, Kubernetes/k3s, Kueue and agent-driven research campaigns.

## Work Experience

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### Axiotic AI — Chief Science Officer / Head of Research

November 2025–present

- Developed the **DAEDALUS multimodal world-model series**, spanning **0.5B, 1B, 4B, 8B, 16B, 32B and 72B parameters**, with persistent memory and latent reasoning. Designed its self-improvement flywheel around user interaction, autonomous agent testing and Axiotic's Byzantine validation methods.
- Trained the series' **72B model from scratch on approximately 1T tokens** in 2026 using **64 H100/B100 GPUs on GCP** (PyTorch, Accelerate, FSDP); owned run design, training performance and execution.
- Designed and ran CHRONOS training experiments on richer temporal learning objectives, personally training the model ladder to **536M parameters and 5.5B tokens on a single RTX 4090**.

### Pieces for Developers — Principal Research Scientist & Head of ML Research

March 2025–March 2026

- Conceived, designed, built and trained **20–80M-parameter language and embedding models**; analysed failures, iterated and optimised them through to working on-device deployment.
- Developed retrieval, memory, evolutionary deep-research methods and large-context small models, connecting model design with evaluation and device constraints. Confidential results omitted.

### Malted AI — Senior Machine Learning Research Scientist / Engineer

May 2024–February 2025

- Built experimentation frameworks and trained/adapted **Llama 3.1, Phi 3.5 mini and Qwen2** with LoRA, QLoRA and full fine-tuning, including small-data settings with **32–2,048 examples**.
- Built teacher-committee synthetic-data pipelines, agreement-based selection and **8B-to-sub-600M distillation** experiments.
- Conceived and built an evolutionary prompt-optimisation system for an 8B model in one day. Results and recipes remain confidential.

## University of Edinburgh — Research Associate; PhD research

Research Associate, 2021–2024 · PhD, 2017–2021

- Developed **MAML++**, **learned self-critique**, unsupervised few-shot methods and **Data Augmentation GANs**: implementation, training and analysis of adaptation and transferable augmentation.
- Built **GATE/EEVEE** multimodal evaluation tooling, **TALI** audio–language–image/video pipelines and research-group Kubernetes infrastructure; authored **kubejobs**.

**Google — Research Intern | 2020–2021**: investigated few-shot transfer under extreme domain shift.

**Amazon — Speech-Scientist Intern | 2016**: speech-science research for Echo. Earlier embedded-hardware design/programming at Lancaster.

## Selected Projects

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- **Research infrastructure**: built Axiotic’s Olympus GKE platform with Kueue, Spot workers, Workload Identity and immutable GCS storage; integrated agents and campaign control. Developed pinned-source execution, checkpoint/resume and W&B/Hugging Face tracking. Built home GPU hardware from scratch: **two nodes, 4× RTX 4090 and 1× RTX 5090**, running k3s/Kueue.
- **Adamantium — Axiotic**: designed and executed BPE-transformer, byte-convolutional and hybrid-embedding campaigns; **5.8–9.4M parameters, approximately 10B teacher-token exposure per run across repeated corpus passes**. Built screening/full evaluation and investigated benchmark/provenance failures.
- **Cliphoard — independent**: built Swift/macOS semantic search using **Ogma models** I trained at Axiotic, deployed through **CoreML**; encrypted storage, model installation and tokenizer/CoreML–PyTorch parity tests.
- **HephaestusZero — independent**: conceived, designed and built an **information-theoretic Byzantine** agent runtime with uncertainty-scaled redundancy, diverse reasoning paths and adversarial verification. Built persistent, multi-resolution memory formation and agentic retrieval: standing context, query-specific exact/tag/BM25 search and context-budgeted selection.
- **MLP & MSc research — Edinburgh**: conceived and pitched projects; developed ideas with student groups through architecture design, experiments, supplied code and training diagnosis. **World models**: CarRacing model-based RL, DARLA/CoinRun disentanglement and transfer. **Generation**: CLIP-conditioned AttnGAN, relational CycleGAN, low-data CLIP/StyleGAN2. **Data-efficient learning**: EvolutionAugment, adversarial augmentation and loss-driven class-imbalance curricula.
- **LeRobot Edinburgh — 2025**: initiated the hackathon, supplied robotics/GPU hardware and mentored teams.

## Education

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**PhD, Machine Learning**, Edinburgh, 2017–2021 · **MScR, Data Science (Distinction)**, Edinburgh, 2016–2017 · **MSc, Data Science (Distinction)**, Lancaster, 2014–2015 · **BEng (Hons), Computer Systems Engineering**, Lancaster, 2011–2014.

## Selected Publications

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How to Train Your MAML (ICLR 2019); Self-Critique (NeurIPS 2019); Contrastive Meta-Learning (ICLR 2023); einspace (NeurIPS 2024).

More available on my [personal website](#) or [Google Scholar page](#).